

# Is a Fault an Initial Condition?

Study F — separating what a fault does to the *state* from what it does to the *vehicle*

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## 1 The question this study exists to answer

The proposal under test, stated as it was put:

*“Computing a nominal trajectory and then injecting a fault seems unnecessary. A fault only changes our system’s and control’s initial states. If we cover most of the initial-condition space, we are good.”*

This is a claim about **what a fault is**, and it is worth taking seriously rather than deferring to convention, because half of it is right and that half is the more useful half.

### 1.1 The half that is right

The dynamics are Markovian. What the vehicle can do from a state  $x$  depends on  $x$  and on the vehicle, and never on the history that produced  $x$ . A vehicle that arrived at a tumbling, gimbal-deflected state by flying a nominal descent and then losing an engine is in *exactly* the same position as one that simply started there. There is no hidden memory in the model for a trajectory to deposit.

So the nominal-trajectory scaffolding is not needed **to generate post-fault states**. Those states can be sampled directly, which is strictly better: it reaches states no single nominal trajectory would ever pass through, and it spends its compute on coverage instead of on re-simulating the same descent. Studies A–E could only reach a post-fault state by simulating forwards into it, which is why between them they examined a few hundred such states, all descended from **one** nominal trajectory.

This study therefore does what was proposed: it drops the nominal trajectory, samples the **full 22-dimensional state** — 12 rigid-body dimensions *and* the 10 actuator states, which is the “and control states” part of the claim — and asks from where a landing is possible.

## 1.2 The half that has to be measured

A fault does something else as well. It changes **the vehicle**, and it keeps it changed for the whole remaining flight. After the fault, engine 2 delivers  $\eta T$  instead of  $T$ , or its gimbal actuator responds at  $\omega_n = 0.6$  rad/s instead of 4.0, or it delivers nothing at all. That is not a value of the state vector. It is a different  $f(x, u)$  and, for a dead engine, a smaller set of admissible controls.

Whether that distinction *matters in practice* is an empirical question, and the answer is not obvious in advance. It is conceivable that a damaged vehicle started from a good state does about as well as a healthy one — in which case the proposal would be right in full, and fault modelling could be replaced by initial-condition dispersion. This study is built to measure exactly that, and Section 4 reports what it found.

# 2 Method

## 2.1 The design that settles it: paired samples

One set of initial conditions, drawn once. **Every plant configuration is solved from the same initial conditions.**

That pairing is the whole design. Because the states are held identical across configurations, any difference in landing rate between two configurations is attributable to the plant alone — the state distribution cancels exactly rather than approximately. If a fault were expressible as an initial condition, then feeding the same states to a healthy vehicle and to a damaged one would produce the same landing set, and every paired difference would be zero.

The statistical test is **McNemar’s exact test**, which uses only the *discordant* pairs: samples where one plant lands and the other does not. Under the null hypothesis that the plant change has no effect, each discordant pair is an independent coin flip, so the two-sided  $p$ -value is an exact binomial test. Pairing also buys precision — it detects a plant effect with far fewer solves than comparing two independently-sampled proportions would need.

## 2.2 The 22-dimensional state box

The 12 rigid dimensions are Study E’s box with the **body-rate range widened from  $\pm 4$  to  $\pm 25$  °/s**. That widening is not cosmetic: Study D measured what a fault actually does to the body rates before anyone reacts (12 to 360 °/s within one second), so a box that stops at 4 °/s cannot contain a post-fault state at all.

The 10 actuator dimensions are new here, and they are the part of the proposal that earlier studies could not test. Per engine: thrust anywhere in its envelope, both gimbal angles anywhere in  $\pm 6^\circ$ , and both gimbal rates. A fault leaves gimbals deflected and thrust away from trim; sampling those states directly reaches them without simulating a descent to get there.

A sample is rejected only if it is **already lost** (tumbling past the hard limits, past horizontal, or on the ground) or outside the approach cone. States far outside the *planner’s* comfort envelope are deliberately kept, because those are precisely the states a fault produces and the study exists to find out which of them are recoverable.

## 2.3 The recovery planner

The algorithmic deliverable is `fault_lib.recover(x, plant, cfg)`. It takes the current 22-state and the vehicle the fault has left behind, and returns a landing trajectory or an honest refusal. It takes **no** nominal trajectory, no fault onset time, and no fault history — by the Markov argument above, it does not need them.

Two mechanisms make it work on states that are far from trim:

- a **recovery corridor** — a state already outside the planner’s attitude/rate envelope has its bound relaxed to the initial excursion and tightened linearly back to the design envelope over six seconds. The planner must demonstrably recover into the envelope; it is a demand, not a dispensation.
- a **state-dependent horizon** — with no nominal there is no “time already flown”, so the horizon is set from the altitude to be lost. Too long is worse than too short: it forces the planner to hold the 1 m contact floor for the unused tail, which Study D established turns feasible problems infeasible.

## 2.4 An integration bug this study exposed

Study F’s first run returned **0 of 12 landings for the healthy control** — a vehicle with nothing wrong with it, failing from every state. A healthy control that cannot land is a bug, not a result, so the campaign was stopped and the cause traced by ablation.

Zeroing the sampled **gimbal rate** states fixed it completely; halving them did not; thrust asymmetry was irrelevant. That pointed at the integrator. The gimbal actuator is a second-order mode at  $\omega_n = 4$  rad/s, and on the 1 s control grid  $\omega_n \Delta t = 4.0$ , which is **beyond RK4’s stability limit of  $\approx 2.79$**  for an oscillatory eigenvalue. A single RK4 step per control interval was amplifying that mode instead of damping it.

Every earlier study started the gimbal rate states at zero, so the unstable mode was never excited and the defect stayed hidden. Study F samples those states directly, which is exactly the condition that exposes it.

The fix is to integrate each control interval in **two RK4 sub-steps** ( $\omega_n \Delta t_{sub} = 2.0$ , comfortably stable), leaving the decision variables and the control grid untouched. Measured on the probe states:

| RK4 sub-steps per interval | healthy control solved | median solve | peak memory |
|----------------------------|------------------------|--------------|-------------|
| 1 (as in Studies A–E)      | 0 / 3                  | 19 s         | 0.63 GB     |
| 2 (adopted)                | 3 / 3                  | 20 s         | 1.46 GB     |
| 4                          | 3 / 3                  | 39 s         | 3.07 GB     |

Two sub-steps fix the problem completely at no cost in solve time. This is worth flagging beyond Study F: **the 1 s grid under-resolves the gimbal actuator in the base model**, and any result that depends on gimbal transients inherits that.

## 3 What was run

| quantity             | value |
|----------------------|-------|
| initial conditions   | 48    |
| plant configurations | 12    |
| total NLP solves     | 576   |

Three fault families — engine-out, thrust efficiency, gimbal degradation — each at two engine spacings.  $y_{eng} = 1.5$  m is the design value;  $y_{eng} = \mathbf{0.25\ m}$  sits just inside Study A’s analytic trim limit  $y_{crit} = d_{z,eng} \tan \delta_{max} = 0.263$  m, so the gimbal alone can trim a one-engine-out asymmetry. The pair separates *loss of thrust* from

*loss of roll authority*: at 0.25 m the vehicle loses exactly the same thrust and keeps the ability to trim it, so anything that survives at 0.25 m but not at 1.5 m is a roll-authority failure, not a thrust failure.

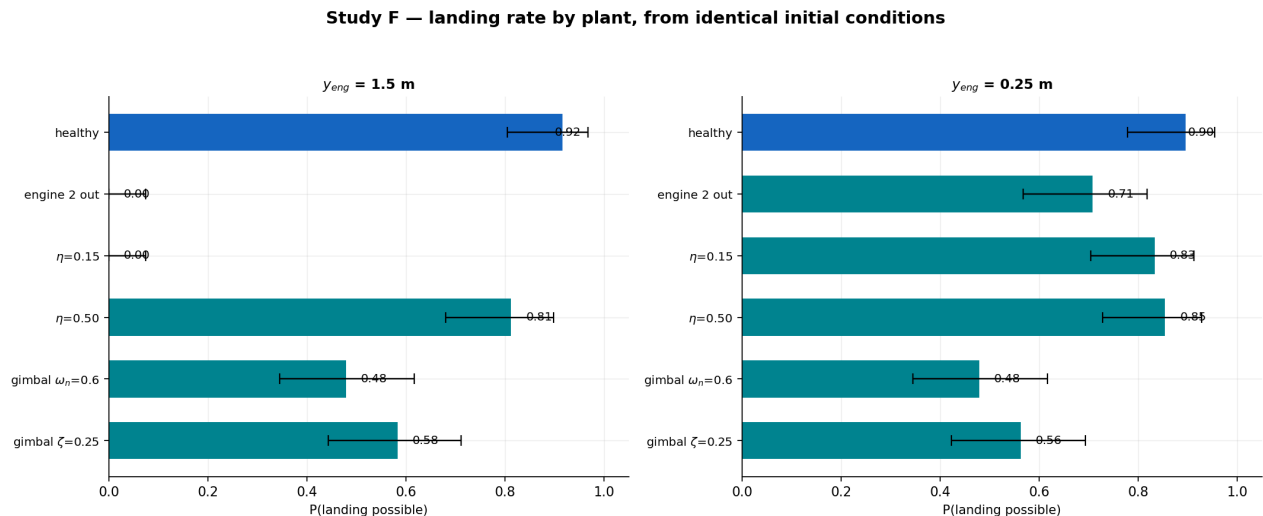


Figure 1: Landing rate by plant

| plant                                    | lands | P(landing possible), 95 % Wilson |
|------------------------------------------|-------|----------------------------------|
| engine 2 out, $y_{eng}=1.5$ m            | 0/48  | <b>0.000</b> [0.000, 0.074]      |
| $\eta=0.15$ , $y_{eng}=1.5$ m            | 0/48  | <b>0.000</b> [0.000, 0.074]      |
| $\eta=0.50$ , $y_{eng}=1.5$ m            | 39/48 | <b>0.812</b> [0.681, 0.898]      |
| gimbal $\omega_n=0.6$ , $y_{eng}=1.5$ m  | 23/48 | <b>0.479</b> [0.345, 0.617]      |
| gimbal $\zeta=0.25$ , $y_{eng}=1.5$ m    | 28/48 | <b>0.583</b> [0.443, 0.712]      |
| healthy, $y_{eng}=1.5$ m                 | 44/48 | <b>0.917</b> [0.804, 0.967]      |
| engine 2 out, $y_{eng}=0.25$ m           | 34/48 | <b>0.708</b> [0.568, 0.818]      |
| $\eta=0.15$ , $y_{eng}=0.25$ m           | 40/48 | <b>0.833</b> [0.704, 0.913]      |
| $\eta=0.50$ , $y_{eng}=0.25$ m           | 41/48 | <b>0.854</b> [0.728, 0.928]      |
| gimbal $\omega_n=0.6$ , $y_{eng}=0.25$ m | 23/48 | <b>0.479</b> [0.345, 0.617]      |
| gimbal $\zeta=0.25$ , $y_{eng}=0.25$ m   | 27/48 | <b>0.562</b> [0.423, 0.693]      |
| healthy, $y_{eng}=0.25$ m                | 43/48 | <b>0.896</b> [0.778, 0.955]      |

## 4 The verdict on the claim

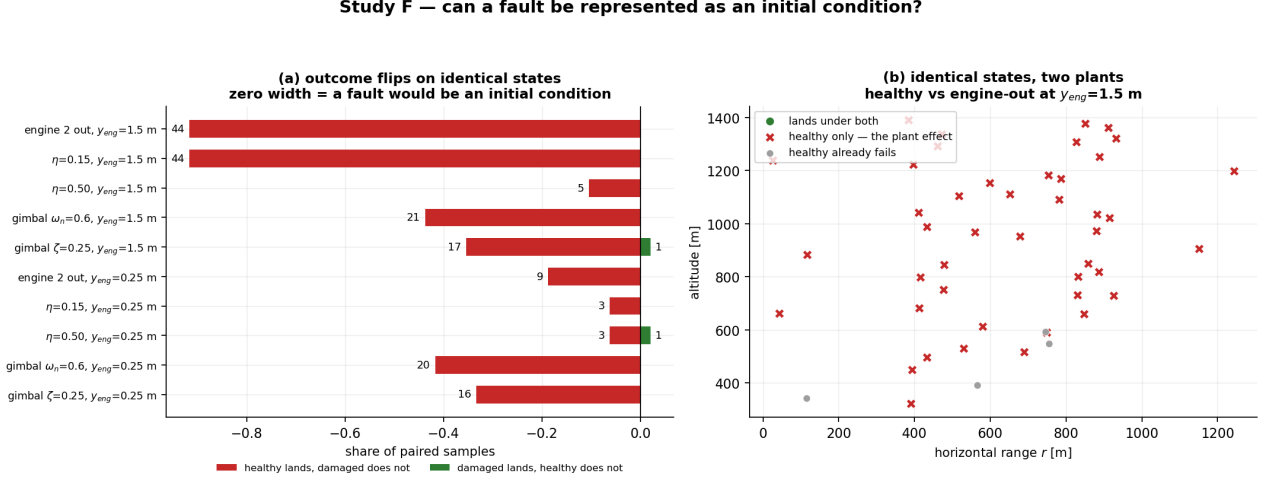


Figure 2: Is a fault an initial condition?

Across all damaged configurations, **480 paired comparisons** on identical initial conditions produced **184 discordant pairs**: 182 where the healthy vehicle lands and the damaged one cannot, and 2 the other way round.

| damaged<br>plant | $y_{eng}$ | healthy only | damaged only | McNemar       | effect? |
|------------------|-----------|--------------|--------------|---------------|---------|
| engine 2 out     | 1.5       | 44           | 0            | $p < 10^{-4}$ | yes     |
| $\eta=0.15$      | 1.5       | 44           | 0            | $p < 10^{-4}$ | yes     |
| $\eta=0.50$      | 1.5       | 5            | 0            | $p = 0.0625$  | no      |
| gimbal           | 1.5       | 21           | 0            | $p < 10^{-4}$ | yes     |
| $\omega_n=0.6$   |           |              |              |               |         |
| gimbal           | 1.5       | 17           | 1            | $p = 0.0001$  | yes     |
| $\zeta=0.25$     |           |              |              |               |         |
| engine 2 out     | 0.25      | 9            | 0            | $p = 0.0039$  | yes     |
| $\eta=0.15$      | 0.25      | 3            | 0            | $p = 0.2500$  | no      |
| $\eta=0.50$      | 0.25      | 3            | 1            | $p = 0.6250$  | no      |
| gimbal           | 0.25      | 20           | 0            | $p < 10^{-4}$ | yes     |
| $\omega_n=0.6$   |           |              |              |               |         |
| gimbal           | 0.25      | 16           | 0            | $p < 10^{-4}$ | yes     |
| $\zeta=0.25$     |           |              |              |               |         |

All rows are 48 paired samples. “healthy only” counts states the healthy vehicle lands from and the damaged one cannot; “damaged only” is the reverse.

### 4.1 The answer is: it depends on whether the vehicle can trim the fault

7 of 10 configurations differ from the healthy vehicle at  $p < 0.05$  on states that are literally identical. The state distribution is not merely similar between the two arms of each comparison — it is the same list of numbers — so those differences cannot be attributed to sampling and can only come from the plant.

But the 3 that are *not* significant are the interesting ones, and they are not a random subset. They are exactly the faults the vehicle has the authority to trim:

| plant       | $y_{eng}$ | discordant (healthy / damaged) | McNemar      |
|-------------|-----------|--------------------------------|--------------|
| $\eta=0.50$ | 1.5       | 5 / 0                          | $p = 0.0625$ |
| $\eta=0.50$ | 0.25      | 3 / 1                          | $p = 0.6250$ |
| $\eta=0.15$ | 0.25      | 3 / 0                          | $p = 0.2500$ |

$\eta = 0.50$  sits above Study C’s saturation limit  $\eta_{sat} = 0.278$ , so the healthy engine can be throttled up to match its partner and the asymmetry is cancelled outright.  $\eta = 0.15$  is far below it — but at  $y_{eng} = 0.25$  m the residual roll moment is inside what the gimbal alone can trim. **In all three cases the vehicle can null the fault, and the outcome reverts to being decided by the state.**

So the sharpened verdict is:

**A fault is not an initial condition — except when the vehicle has enough control authority to cancel it, in which case it very nearly is.**

That is a real and useful vindication of the intuition, with a stated boundary. Where the fault is trimmable, modelling it as a state perturbation is defensible and the plant change is statistically undetectable over 48 paired samples. Where it is not trimmable, the plant dominates so completely that no coverage of the state space can substitute for it: at  $y_{eng} = 1.5$  m the engine-out and  $\eta = 0.15$  vehicles land from **zero** of the 48 states, while the healthy vehicle lands from 44 of the same 48.

## 4.2 The landing sets are nested, not merely different

Across every configuration, the damaged vehicle lands from a state where the healthy one fails at most 1 time. The discordance is essentially one-directional: damage **strictly shrinks** the recoverable set rather than moving it somewhere else.

This matters for the proposal. If damage merely *displaced* the landing set, then sampling initial conditions widely enough really would cover the post-fault cases — you would just be looking at a different part of the same box. Because the set shrinks instead, there is no region of state space that a damaged vehicle handles and a healthy one does not, and therefore no amount of state-space coverage that recovers the missing information about  $f$ .

## 4.3 What this means for the approach

The right synthesis keeps both halves, and it is what this study implements:

1. **Drop the nominal trajectory** — as proposed. It is not needed to reach post-fault states, and sampling the 22-D state directly covers far more of the space for the same compute. This part of the intuition was correct, and Studies A–E were doing unnecessary work.
2. **Keep the fault as a plant change** — sweep it as a *parameter of the dynamics*, crossed with the state box rather than folded into it. For trimmable faults this costs almost nothing (the two collapse); for untrimmable ones it is the entire answer.

That is exactly the “(plant)  $\times$  (initial state)” grid run here, and it answers “when can we land and when can we not?” for each fault family separately, without ever computing a nominal descent.

It is worth being precise about where this leaves the supervisor’s advice. “Compute the nominal, then inject” is not wrong, but the usual justification for it is: it is a *convenient* way to generate plausible post-fault states, not a necessary one. Its real weakness is that every state it produces descends from a single trajectory, which is a thin and unrepresentative slice of the space — this study reached 48 independent states per configuration, none of which any one nominal descent would have visited. Sampling directly is better on that axis. What the nominal-injection framing must not be used to discard is the plant change, which for untrimmable faults is the whole of the answer.

## 5 Which faults are survivable, and why

Study F — failure modes and the engine-spacing contrast

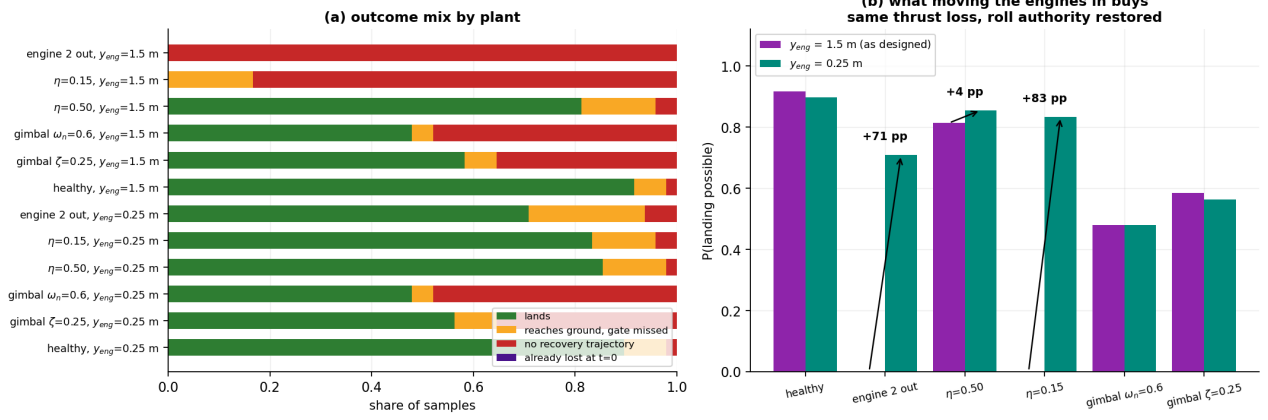


Figure 3: Failure modes and the engine-spacing contrast

### 5.1 Moving the engines in

| plant                 | $y_{eng} = 1.5$ m | $y_{eng} = 0.25$ m | change |
|-----------------------|-------------------|--------------------|--------|
| healthy               | 0.917             | 0.896              | -2 pp  |
| engine 2 out          | 0.000             | 0.708              | +71 pp |
| $\eta=0.50$           | 0.812             | 0.854              | +4 pp  |
| $\eta=0.15$           | 0.000             | 0.833              | +83 pp |
| gimbal $\omega_n=0.6$ | 0.479             | 0.479              | +0 pp  |
| gimbal $\zeta=0.25$   | 0.583             | 0.562              | -2 pp  |

The largest gain is  $\eta=0.15$ , at +83 percentage points. The vehicle at 0.25 m loses *exactly the same thrust* as at 1.5 m — same engine, same  $\eta$ , same total force — so any improvement is attributable purely to roll authority.

This isolates the mechanism that Studies A and D reached from the other direction. Study A’s static budget says one dead engine at 1.5 m applies 18 796 N·m of roll against 5 432 N·m of combined gimbal and RCS authority. At 0.25 m the same dead engine applies 3 133 N·m, which the gimbal can trim on its own. The thrust loss is identical and survivable; the roll asymmetry is what was killing the vehicle.

### 5.2 Three fault families, three different mechanisms

The table above separates the three families cleanly, and they do not behave alike:

- **Asymmetry faults** (engine-out,  $\eta = 0.15$ ) — fatal at 1.5 m, largely survivable at 0.25 m. Moving the engines in converts them from unrecoverable to routine, because the fault was never about thrust.
- **Trimable thrust loss** ( $\eta = 0.50$ , above  $\eta_{sat}$ ) — survivable at both spacings and barely distinguishable from healthy. The spacing change buys almost nothing (+4 pp) because there was nothing to fix.
- **Gimbal degradation** — costs 34 to 44 points of landing probability and is **completely indifferent to the spacing** (0.479 to 0.479 for the sluggish actuator). It is not an asymmetry fault at all: the engines still balance, and what is lost is *bandwidth* — the ability to respond quickly enough to the disturbances the descent itself generates. No amount of moving the engines recovers a slow actuator.

The healthy control is the check on all of this: it lands from 0.917 of states at 1.5 m and 0.896 at 0.25 m — the same within sampling error. The spacing change is not a general improvement to the vehicle; it does nothing for a healthy one, and everything for an asymmetrically damaged one. **That specificity is what identifies roll authority, rather than anything else, as the mechanism.**

The design consequence is the same one Study A reached and is now confirmed from an independent direction: the 1.5 m spacing is simultaneously the vehicle’s best roll actuator and its single point of failure, and it is worth exactly nothing when the vehicle is healthy.

### 5.3 Which parts of the state space are recoverable

A classifier fitted to the *damaged* plants only, over all 22 state dimensions, answers “given something is broken, where in the state space can it still be saved?”.

**Engine spacing y0.25** (165/240 land, out-of-fold accuracy 0.758):

| state dimension | permutation importance |
|-----------------|------------------------|
| alt             | 0.1177                 |
| w               | 0.0319                 |
| u               | 0.0253                 |
| r               | 0.0170                 |
| dp1_dot         | 0.0080                 |
| phi             | 0.0073                 |

**Engine spacing y1.5** (90/240 land, out-of-fold accuracy 0.500):

| state dimension | permutation importance |
|-----------------|------------------------|
| w               | 0.0378                 |
| u               | 0.0250                 |
| alt             | 0.0229                 |
| dy2_dot         | 0.0174                 |
| r               | 0.0156                 |
| theta           | 0.0153                 |

This contrast is itself the study’s thesis restated. A classifier that sees **only the state** — all 22 dimensions of it — and not which plant is flying scores 0.500 at  $y_{eng} = 1.5$  m and 0.758 at 0.25 m.

At the design spacing the state carries essentially no information about the outcome, because the outcome is decided almost entirely by *which fault* the vehicle has: engine-out and  $\eta = 0.15$  land from nowhere,  $\eta = 0.50$  lands from almost everywhere, and no coordinate of  $x$  changes that. At 0.25 m, where every fault is trimmable, the plant stops dominating and the state becomes informative again — the classifier recovers to 0.758, and the dimensions it leans on (altitude, vertical and forward speed) are the ordinary energy-management ones.

**A state-only model of a fault is uninformative exactly where the fault matters most.** That is the quantitative form of the answer to the original question.

## 6 What this study does not say

- **Open-loop replanning, not closed-loop control.** *recover* solves a fresh optimal-control problem that knows the plant exactly. A real vehicle flies a feedback law and must *identify* the fault first. These results are an optimistic bound on what any real guidance system could achieve.
- **The plant is known at the moment of recovery.** Fault detection and identification are not modelled. In practice the plant estimate arrives late and wrong, and both shrink the numbers here.



- **A box is not a distribution.** The state box has uniform density and no covariance structure. Real post-fault states are concentrated on a thin manifold inside it — Study D’s states, for instance — so these fractions describe the box, not the operational likelihood of landing.
- **One fault at a time, on engine 2.** The vehicle is laterally symmetric so the engine index does not matter, but simultaneous or cascading faults are not covered.
- **Local infeasibility is not global.** The NLP is nonconvex; `no_recovery` means IPOPT failed from two seeds, not that no trajectory exists.
- **Constant mass,** inherited from the base model, which understates the propellant cost of a partial fault (Study C).